Lab Assignment 05

BRAC UNIVERSITY

Inspiring Excellence

Course Code:	**CSE111**
Course Title:	**Programming Language II**
Topic:	**Multi-class Design**
Number of Tasks:	**10**

[You are not allowed to change the driver codes of any of the tasks]

After YouTube Music, Spotify has decided to redesign their Playlist system. However, they decided to **not use arrays** to store their music, instead, they will use OOP concepts to create the new Playlist system. You have been assigned to build the system by using 3 classes (**Song**, **Playlist,** and **SpotifyTester**).
Each song will have the *name of the song*, *artist name*, *length of the song in minutes* and *the next song*. Each playlist will have a name and it can contain multiple songs. Both classes will have some features which will be demonstrated in each task.
[You are not allowed to use Array or any built-in libraries for this assignment]

Task 1

Design the **Song** class with *constructor* and *songInfo()* method along with necessary attributes in such a way that it produces the following output.

Driver Code	Output

`public class SpotifyTester {`
` public static void main(String[] args) {`
`  Song s1 = new Song("Song-A", "Artist-A", 3);`
`  System.out.println("1==========");`
`  s1.songInfo();`
`  System.out.println("2==========");`
`//  More lines will be added in this Tester class`
	`1==========`
	`Title: Song-A`
	`Artist: Artist-A`
	`Length: 3 minutes`
	`2==========`

Task 2

Design the **Playlist** class constructor along with necessary attributes in such a way that it produces the following output.

Driver Code	Output

`  System.out.println("2==========");`
`//  Continuation from Task 1`
`  Playlist p1 = new Playlist("First Playlist");`
`  System.out.println("3==========");`
	`2==========`
	`First Playlist created.`
	`3==========`

Task 3 & 4

Create *addSong()* method and *info()* method inside the **Playlist** class to produce the following output.

Driver Code	Output

`System.out.println("3==========");`
`//  Continuation from Task 2`
`p1.info();`
`System.out.println("4==========");`
`p1.addSong(s1);`
`System.out.println("5==========");`
`p1.info();`
`System.out.println("6==========");`
`Song s2 = new Song("Song-B", "Artist-B", 4);`
`Song s3 = new Song("Song-C", "Artist-C", 2);`
`p1.addSong(s2);`
`p1.addSong(s3);`
`System.out.println("7==========");`
`p1.info();`
`System.out.println("8==========");`
	`3==========`
	`First Playlist has the following songs:`
	`No songs in First Playlist.`
	`4==========`
	`Song-A added to First Playlist.`
	`5==========`
	`First Playlist has the following songs:`
	`Song-1`
	`Title: Song-A`
	`Artist: Artist-A`
	`Length: 3 minutes`
	`6==========`
	`Song-B added to First Playlist.`
	`Song-C added to First Playlist.`
	`7==========`
	`First Playlist has the following songs:`
	`Song-1`
	`Title: Song-A`
	`Artist: Artist-A`
	`Length: 3 minutes`
	`Song-2`
	`Title: Song-B`
	`Artist: Artist-B`
	`Length: 4 minutes`
	`Song-3`
	`Title: Song-C`
	`Artist: Artist-C`
	`Length: 2 minutes`
	`8==========`

Task 5

Create *addSong()* [overloaded] method inside the **Playlist** class to produce the following output.

Driver Code	Output

`System.out.println("8==========");`
`//  Continuation from Task 3&4`
`Song s4 = new Song("Song-D", "Artist-D", 3);`
`Song s5 = new Song("Song-E", "Artist-E", 4);`
`Song s6 = new Song("Song-F", "Artist-F", 2);`
`Song s7 = new Song("Song-G", "Artist-G", 2);`
`p1.addSong(s4, 0);`
`p1.addSong(s5, 2);`
`p1.addSong(s6, 5);`
`p1.addSong(s7, 10);`
`System.out.println("9==========");`
`p1.info();`
`System.out.println("10==========");`
	`8==========`
	`Song-D added to First Playlist.`
	`Song-E added to First Playlist.`
	`Song-F added to First Playlist.`
	`Cannot add song to Index 10.`
	`9==========`
	`First Playlist has the following songs:`
	`Song-1`
	`Title: Song-D`
	`Artist: Artist-D`
	`Length: 3 minutes`
	`Song-2`
	`Title: Song-A`
	`Artist: Artist-A`
	`Length: 3 minutes`
	`Song-3`
	`Title: Song-E`
	`Artist: Artist-E`
	`Length: 4 minutes`
	`Song-4`
	`Title: Song-B`
	`Artist: Artist-B`
	`Length: 4 minutes`
	`Song-5`
	`Title: Song-C`
	`Artist: Artist-C`
	`Length: 2 minutes`
	`Song-6`
	`Title: Song-F`
	`Artist: Artist-F`
	`Length: 2 minutes`
	`10==========`

Task 6

Create *playSong()* method inside the **Playlist** class to produce the following output.

Driver Code	Output

`System.out.println("10==========");`
`//  Continuation from Task 5`
`p1.playSong("Song-F");`
`p1.playSong("Song-G");`
`p1.playSong("Song-B");`
`System.out.println("11==========");`
	`10==========`
	`Playing Song-F by Artist-F.`
	`Song-G not found in playlist First Playlist.`
	`Playing Song-B by Artist-B.`
	`11==========`

Task 7

Create the *playSong()* [overloaded] method inside the **Playlist** class to produce the following output.

Driver Code	Output

`System.out.println("11==========");`
`//  Continuation from Task 6`
`p1.playSong(0);`
`p1.playSong(4);`
`p1.playSong(8);`
`System.out.println("12==========");`
	`11==========`
	`Playing Song-D by Artist-D.`
	`Playing Song-C by Artist-C.`
	`Song at Index 8 not found in First Playlist.`
	`12==========`

Task 8

Create the *deleteSong()* method inside the **Playlist** class to produce the following output.

Driver Code	Output

`System.out.println("12==========");`
`//  Continuation from Task 7`
`p1.deleteSong("Song-D");`
`p1.deleteSong("Song-B");`
`p1.deleteSong("Song-F");`
`p1.deleteSong("Song-K");`
`System.out.println("13==========");`
`p1.info();`
`System.out.println("14==========");`
	`12==========`
	`Song-D deleted from First Playlist.`
	`Song-B deleted from First Playlist.`
	`Song-F deleted from First Playlist.`
	`Song-K not found in First Playlist.`
	`13==========`
	`First Playlist has the following`
	`songs:`
	`Song-1`
	`Title: Song-A`
	`Artist: Artist-A`
	`Length: 3 minutes`
	`Song-2`
	`Title: Song-E`
	`Artist: Artist-E`
	`Length: 4 minutes`
	`Song-3`
	`Title: Song-C`
	`Artist: Artist-C`
	`Length: 2 minutes`
	`14==========`

Task 9

Create the *totalSong()* method inside the **Playlist** class to produce the following output.

Driver Code	Output

`System.out.println("14==========");`
`//  Continuation from Task 8`
`System.out.println(p1.name +" has "+p1.totalSong()`
`+" songs");`
`System.out.println("15==========");`
	`14==========`
	`First Playlist has 3 songs`
	`15==========`

Task 10

Create the *merge()* method inside the **Playlist** class to produce the following output.

Driver Code	Output

`System.out.println("15==========");`
`//  Continuation from Task 9`
`Song ns1 = new Song("Song-Z", "Artist-Z", 3);`
`Song ns2 = new Song("Song-Y", "Artist-Y", 4);`
`Song ns3 = new Song("Song-X", "Artist-X", 2);`
`System.out.println("16==========");`
`Playlist p2 = new Playlist("Second Playlist");`
`p2.addSong(ns1);`
`p2.addSong(ns2);`
`p2.addSong(ns3);`
`System.out.println("17==========");`
`p1.info();`
`System.out.println("18==========");`
`p2.info();`
`System.out.println("19==========");`
`p1.merge(p2);`
`System.out.println("20==========");`
`p1.info();`
`System.out.println("21==========");`
	`15==========`
	`16==========`
	`Second Playlist created.`
	`Song-Z added to Second Playlist.`
	`Song-Y added to Second Playlist.`
	`Song-X added to Second Playlist.`
	`17==========`
	`First Playlist has the following songs:`
	`Song-1`
	`Title: Song-A`
	`Artist: Artist-A`
	`Length: 3 minutes`
	`Song-2`
	`Title: Song-E`
	`Artist: Artist-E`
	`Length: 4 minutes`
	`Song-3`
	`Title: Song-C`
	`Artist: Artist-C`
	`Length: 2 minutes`
	`18==========`
	`Second Playlist has the following songs:`
	`Song-1`
	`Title: Song-Z`
	`Artist: Artist-Z`
	`Length: 3 minutes`
	`Song-2`
	`Title: Song-Y`
	`Artist: Artist-Y`
	`Length: 4 minutes`
	`Song-3`
	`Title: Song-X`
	`Artist: Artist-X`
	`Length: 2 minutes`
	`19==========`
	`Merge Completed!`
	`20==========`
	`First Playlist has the following songs:`
	`Song-1`
	`Title: Song-A`

	Artist: Artist-A Length: 3 minutes Song-2 Title: Song-E Artist: Artist-E Length: 4 minutes Song-3 Title: Song-C Artist: Artist-C Length: 2 minutes Song-4 Title: Song-Z Artist: Artist-Z Length: 3 minutes Song-5 Title: Song-Y Artist: Artist-Y Length: 4 minutes Song-6 Title: Song-X Artist: Artist-X Length: 2 minutes 21==========

Task 11 [Ungraded]

Create the *showHistory()* method inside the **Playlist** class to produce the following output. [Hint: showHistory() only shows the songs which were played from the playlist. So you might need to update the method which is used to play Songs.]

Driver Code	Output

`System.out.println("21==========");`
`//  Continuation from Task 10`
`p1.showHistory();`
`System.out.println("22==========");`
`p2.showHistory();`
`System.out.println("23==========");`
`}`
`}`
	`21==========`
	`History of First Playlist:`
	`Title: Song-F`
	`Artist: Artist-F`
	`Length: 2 minutes`
	`Title: Song-B`
	`Artist: Artist-B`
	`Length: 4 minutes`
	`Title: Song-D`
	`Artist: Artist-D`
	`Length: 3 minutes`
	`Title: Song-C`
	`Artist: Artist-C`
	`Length: 2 minutes`
	`22==========`
	`History of Second Playlist:`
	`No songs were played from Second Playlist.`
	`23==========`

Complete driver code and expected output:

Driver Code	Output

`public class SpotifyTester {`
` public static void main(String[] args) {`
`  Song s1 = new Song("Song-A", "Artist-A", 3);`
`  System.out.println("1==========");`
`  s1.songInfo();`
`  System.out.println("2==========");`
`  Playlist p1 = new Playlist("First Playlist");`
`  System.out.println("3==========");`
`  p1.info();`
`  System.out.println("4==========");`
`  p1.addSong(s1);`
`  System.out.println("5==========");`
`  p1.info();`
`  System.out.println("6==========");`
`  Song s2 = new Song("Song-B", "Artist-B", 4);`
`  Song s3 = new Song("Song-C", "Artist-C", 2);`
`  p1.addSong(s2);`
`  p1.addSong(s3);`
`  System.out.println("7==========");`
`  p1.info();`
`  System.out.println("8==========");`
`  Song s4 = new Song("Song-D", "Artist-D", 3);`
`  Song s5 = new Song("Song-E", "Artist-E", 4);`
`  Song s6 = new Song("Song-F", "Artist-F", 2);`
`  Song s7 = new Song("Song-G", "Artist-G", 2);`
`  p1.addSong(s4, 0);`
`  p1.addSong(s5, 2);`
`  p1.addSong(s6, 5);`
`  p1.addSong(s7, 10);`
`  System.out.println("9==========");`
`  p1.info();`
`  System.out.println("10==========");`
`  p1.playSong("Song-F");`
`  p1.playSong("Song-G");`
`  p1.playSong("Song-B");`
`  System.out.println("11==========");`
`  p1.playSong(0);`
`  p1.playSong(4);`
`  p1.playSong(8);`
`  System.out.println("12==========");`
`  p1.deleteSong("Song-D");`
`  p1.deleteSong("Song-B");`
`  p1.deleteSong("Song-F");`
`  p1.deleteSong("Song-K");`
	`1==========`
	`Title: Song-A`
	`Artist: Artist-A`
	`Length: 3 minutes`
	`2==========`
	`First Playlist created.`
	`3==========`
	`First Playlist has the following songs:`
	`No songs in First Playlist.`
	`4==========`
	`Song-A added to First Playlist.`
	`5==========`
	`First Playlist has the following songs:`
	`Song-1`
	`Title: Song-A`
	`Artist: Artist-A`
	`Length: 3 minutes`
	`6==========`
	`Song-B added to First Playlist.`
	`Song-C added to First Playlist.`
	`7==========`
	`First Playlist has the following songs:`
	`Song-1`
	`Title: Song-A`
	`Artist: Artist-A`
	`Length: 3 minutes`
	`Song-2`
	`Title: Song-B`
	`Artist: Artist-B`
	`Length: 4 minutes`
	`Song-3`
	`Title: Song-C`
	`Artist: Artist-C`
	`Length: 2 minutes`
	`8==========`
	`Song-D added to First Playlist.`
	`Song-E added to First Playlist.`
	`Song-F added to First Playlist.`
	`Cannot add song to Index 10.`

```java
   System.out.println("13==========");
   p1.info();
   System.out.println("14==========");
   System.out.println(p1.name +" has "+
p1.totalSong() + " songs");
   System.out.println("15==========");
   Song ns1 = new Song("Song-Z", "Artist-Z", 3);
   Song ns2 = new Song("Song-Y", "Artist-Y", 4);
   Song ns3 = new Song("Song-X", "Artist-X", 2);
   System.out.println("16==========");
   Playlist p2 = new Playlist("Second Playlist");
   p2.addSong(ns1);
   p2.addSong(ns2);
   p2.addSong(ns3);
   System.out.println("17==========");
   p1.info();
   System.out.println("18==========");
   p2.info();
   System.out.println("19==========");
   p1.merge(p2);
   System.out.println("20==========");
   p1.info();
   System.out.println("21==========");

//Ungraded Task
   p1.showHistory();
   System.out.println("22==========");
   p2.showHistory();
   System.out.println("23==========");
 }
}
```

```
9===========
First Playlist has the following songs:
Song-1
Title: Song-D
Artist: Artist-D
Length: 3 minutes
Song-2
Title: Song-A
Artist: Artist-A
Length: 3 minutes
Song-3
Title: Song-E
Artist: Artist-E
Length: 4 minutes
Song-4
Title: Song-B
Artist: Artist-B
Length: 4 minutes
Song-5
Title: Song-C
Artist: Artist-C
Length: 2 minutes
Song-6
Title: Song-F
Artist: Artist-F
Length: 2 minutes
10==========
Playing Song-F by Artist-F.
Song-G not found in playlist First
Playlist.
Playing Song-B by Artist-B.
11==========
Playing Song-D by Artist-D.
Playing Song-C by Artist-C.
Song at Index 8 not found in First
Playlist.
12==========
Song-D deleted from First Playlist.
Song-B deleted from First Playlist.
Song-F deleted from First Playlist.
Song-K not found in First Playlist.
13==========
```

	First Playlist has the following songs: Song-1 Title: Song-A Artist: Artist-A Length: 3 minutes Song-2 Title: Song-E Artist: Artist-E Length: 4 minutes Song-3 Title: Song-C Artist: Artist-C Length: 2 minutes 14========== First Playlist has 3 songs 15========== 16========== Second Playlist created. Song-Z added to Second Playlist. Song-Y added to Second Playlist. Song-X added to Second Playlist. 17========== First Playlist has the following songs: Song-1 Title: Song-A Artist: Artist-A Length: 3 minutes Song-2 Title: Song-E Artist: Artist-E Length: 4 minutes Song-3 Title: Song-C Artist: Artist-C Length: 2 minutes 18========== Second Playlist has the following songs: Song-1 Title: Song-Z Artist: Artist-Z Length: 3 minutes

	Song-2 Title: Song-Y Artist: Artist-Y Length: 4 minutes Song-3 Title: Song-X Artist: Artist-X Length: 2 minutes 19========== Merge Completed! 20========== First Playlist has the following songs: Song-1 Title: Song-A Artist: Artist-A Length: 3 minutes Song-2 Title: Song-E Artist: Artist-E Length: 4 minutes Song-3 Title: Song-C Artist: Artist-C Length: 2 minutes Song-4 Title: Song-Z Artist: Artist-Z Length: 3 minutes Song-5 Title: Song-Y Artist: Artist-Y Length: 4 minutes Song-6 Title: Song-X Artist: Artist-X Length: 2 minutes 21========== History of First Playlist: Title: Song-F Artist: Artist-F Length: 2 minutes Title: Song-B

	Artist: Artist-B Length: 4 minutes Title: Song-D Artist: Artist-D Length: 3 minutes Title: Song-C Artist: Artist-C Length: 2 minutes 22========== History of Second Playlist: No songs were played from Second Playlist. 23==========